Umair Javaid Manj

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Research Experience

Doctoral Researcher — **Auckland University of Technology**, Health & Artificial Intelligence, Auckland — Oct 2025 to Present

- Developing an AI based biomarker of chronic low back pain from cortical responses to paraspinal stimulation.
- Extending fMRI mapping of the back (Cole et al., 2022; Dörig et al., 2026) into the EEG dimension, adding millisecond timing and oscillatory dynamics, and comparing painful with healthy spinal segments within the same person.
- Designing a case control EEG study combining source imaging, representational similarity analysis and subject level machine learning.
- Led the first pilot and identified an analysis artefact behind an apparent decoding effect.

Data & AI Governance Analyst — **New Zealand College of Chiropractic**, Auckland — Aug 2026 to Present

- Analyse EEG hyperscanning of chiropractor and patient: inter brain and within brain connectivity (wPLI, dPLI) and source level network analysis.
- Apply graph theory to resting state EEG networks in a randomised chiropractic trial of adults and children, with linear mixed models.
- Model longitudinal clinical outcomes (immune response, DASS-21, PROMIS-25) with mixed effects models in R.
- Manage private participant data and consent records.

Program Manager — **National University of Sciences and Technology**, Islamabad — Sep 2024 to Oct 2025

- Led a national AI training program.

Instructor — **AI Lounge**, Islamabad — May 2023 to Jul 2024

- Supervised 18 student AI projects; awarded Best Instructor.

Research Associate — **COGNET Lab, SEECS**, National University of Sciences and Technology, Islamabad — Aug 2021 to Apr 2023

- Researched the robustness of deep learning and computer vision models to adversarial inputs.
- Mentored junior research assistants (Research Assistant until Nov 2022).

Publications

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. (2026). Application of machine learning techniques in the measurement of ground reaction force (GRF) waveforms using inertial measurement unit (IMU) sensors: a systematic literature review. *Medical Engineering & Physics*. doi
- Rizvi, D. Q., Majeed, M., **Manj, U. J.**, & Haq, M. I. U. (2025). Sleep disorder prediction with PSO-optimized machine and deep learning models. *ICODT2*, pp. 1 to 6. doi
- Haq, M. I. U., Ahmad, Z., **Manj, U. J.**, Akhter, M. N., Arfeen, Z. A., & Qamar, S. D. (2026). Bio-inspired soft robotic gripper with variable stiffness using magneto-rheological control. *ICRAI*, pp. 1 to 6. doi
- Haq, M. I. U., **Manj, U. J.**, Paracha, S. Q., Imam, Y., Rizvi, D. Q., & Farooqi, A. R. (2026). Neuro-symbolic AI-driven secure communication framework for autonomous cyber-physical systems. *APAN 2026*, pp. 1 to 6. doi
- Akhter, M. N., **Manj, U. J.**, Ali, H., Niazi, I. K., & Taylor, D. Deep learning-based kinetics and center of pressure estimation during sit-to-walk transitions from a single sacrum-mounted IMU. *IEEE Transactions on Biomedical Engineering* (under review).

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. Predicting ground reaction force during sit to walk: towards kinetically instrumenting the Timed Up and Go test. *Medical & Biological Engineering & Computing* (under review).
- **Manj, U. J.**, Rizvi, S. D. Q., Niazi, I. K., Akhter, M. N., & Ali, H. Gated feature fusion and age confounding in plantar thermography for diabetes screening. *Biomedical Physics & Engineering Express* (under review).
- **Manj, U. J.**, Qamar, D., Rashid, M. A., Perwaiz, N., & Niazi, I. K. Speaker leakage and subject level accuracy in voice biomarkers for Parkinson's disease. *Machine Learning: Health* (under review).

Education

PhD	**Auckland University of Technology**, Health & Artificial Intelligence, Auckland, New Zealand	Oct 2025 to Present
MS	**National University of Sciences and Technology**, Computational Science & Engineering, Islamabad	2017 to 2021
BE	**National University of Sciences and Technology**, Computer Engineering, Islamabad	2012 to 2016

Skills

Neurophysiology & Signals: EEG preprocessing (ICA, ASR, GEDAI), ERP and time frequency analysis, source localisation, functional connectivity (wPLI, dPLI) and graph theory, hyperscanning, IMU analysis, vibrotactile stimulation

EEG Toolboxes: MNE-Python, autoreject, EEGLAB (ICLabel, ROIconnect), FieldTrip, BrainNet Viewer

Machine Learning: CNN transfer learning (DenseNet, ResNet, ConvNeXt), multi backbone feature fusion with gating and squeeze and excitation, TabNet, TabPFN, transformers and graph neural networks; gradient boosting (XGBoost, LightGBM, CatBoost), SVM, random forests, regularised logistic regression, stacking ensembles

Model Evaluation: nested and subject grouped cross validation, leave one subject out, leakage diagnostics, confounder control, Bayesian optimisation (Optuna, Gaussian processes) and particle swarm optimisation, calibration (ECE, Brier), explainability (SHAP, LIME)

Statistics: hypothesis testing (exact McNemar, Wilcoxon signed rank) with Holm correction, cluster bootstrap and Clopper-Pearson confidence intervals, covariate adjusted AUC, permutation tests, linear mixed models (lme4, lmerTest, emmeans), cluster based permutation for EEG, power analysis (G*Power)

Programming: Python (PyTorch, timm, scikit-learn, NumPy, SciPy, pandas, NetworkX), R, MATLAB, C++; HPC with Slurm

Research Practice: Participant data governance, systematic review, scientific writing